

Request for methodological feedback: a connectivity-mapping pilot that failed its own validation

Not a result. I am asking whether the proposed next test is worth running, and what would make it sound.

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What I did

Scored 5,425 LINCS L1000 **consensus** drug signatures against the CellAge senescence meta-signature (150 genes up / 150 down) and SenMayo, using one-sided Fisher overlap tests, to look for compounds whose expression effect opposes senescence.

What happened

A follow-up validation, specified before scoring but **after** the first results were visible (disclosed — not a blind pre-registration), failed its own criteria:

Test	Result
MDM2/CDK inhibitors on a second senescence gene set (Fridman, no shared genes)	AUC 0.53 / 0.60, Holm $p \geq 0.43$ — no replication
Same, after regressing out a proliferation-arrest score	AUC 0.55 / 0.51, $p = 1.0$ — signal gone
Known senomorphics (mTOR, JAK, p38 inhibitors) on the SASP score	AUC 0.51, $p = 1.0$
Known senolytics (BCL-2, HSP90, Na^+/K^+ -ATPase)	no discrimination

The reversal score correlates **−0.73** with plain proliferation arrest; the SASP score correlates **0.48** with a generic HALLMARK inflammatory-response score. An earlier, circular control-class result (classes chosen after seeing output, AUC 0.01) has been withdrawn. Rankings are stable across 32 scoring configurations, but stability is not validity.

My reading: this pipeline did not demonstrate senescence specificity beyond proliferation arrest. It is an informative negative about this design — consensus signatures, mostly from proliferating cancer lines, scored with set-overlap statistics — not a claim about connectivity mapping in general.

What I am considering next, and where I would like to be corrected

Using GSE329184 (Tan et al., *npj Aging* 2026; NHDF/NHLF fibroblasts, 2 donors, ~230 RNA-seq samples: proliferating, rapamycin quiescence-like, bleomycin / palbociclib / late-passage senescence, ± navitoclax 0.1 and 1 μM), plus GSE162175 from a different lab for replication.

Test A — a senescent-vs-arrested signature (feasible, not yet validated). Build the query from senescent vs arrested cells rather than senescent vs proliferating, then re-run the L1000 screen and repeat the failed validations. In GSE329184 the arrest comparator is rapamycin, so a signature from it alone could separate senescence from *rapamycin's specific effects* rather than from arrest in general. Replication with a **different arrest mechanism** is therefore essential: GSE162175 (Wagner et al., *Nat Commun* 2026; IMR90, 4 replicates) has quiescence by **serum withdrawal** (GEO: 0.5% FBS, days 3–8). Caveats: I found no explicit arrest phenotyping of the sequenced quiescent samples (the paper's imaging

arm used 0.1% FBS, so the conditions may differ), and palbociclib-induced senescence appears in both datasets, so palbociclib must be excluded from any inducer test built on them.

- Q1: Are rapamycin-induced and serum-withdrawal quiescence together an adequate pair of arrest comparators, or is a third mechanism (e.g. contact inhibition) needed before a shared senescence-vs-arrest signature means anything?
- Q2: Is scoring L1000 consensus signatures salvageable at all, or should this move to per-signature querying with cell-line, dose and time as covariates?

Test B — selective senolysis vs general toxicity (paused). I had planned to compare navitoclax in senescent vs matched non-senescent cells. **The currently identified measurements do not support a reliable test of selective senolysis versus general toxicity. We welcome advice on whether additional measurements or supplements make that comparison possible.** The main reason is the lack of an interpretable survival endpoint: there is no viability assay, and the proxy (nuclei count) rises after navitoclax, which the authors themselves read as "supportive of increased mitosis **or** apoptosis"; in one arm it does not shift at all. The readouts are on separate but layout-matched plates ("Technical triplicates were prepared for each assay"). That alone need not rule out a condition-level comparison, if the survival assay, replication and matching were adequate.

- Q3: Is there a dataset with drug-treated senescent and matched non-senescent cells and an **interpretable viability endpoint** (same wells, or properly matched parallel plates) alongside expression? That seems to be the binding constraint, not analysis method.
- Q4: Given that expression is measured only in surviving cells, what design actually distinguishes selective killing from a shift in the surviving population?

Also noted: the same paper reports SenMayo was only partially enriched in chemically induced senescence and not in late-passage cells, while the Hernandez-Segura in-vitro fibroblast signature matched. That is a **coverage** concern (limited sensitivity, or context dependence, in cultured fibroblasts), not by itself a specificity finding. Specificity would need a test of whether SenMayo also responds in non-senescent conditions such as ordinary inflammation. My pilot's SASP score correlated 0.48 with a generic inflammatory-response score, which hints at this but was not a designed specificity test.

What I am asking

Not for collaboration or data. Just: **is Test A worth running, and is Test B answerable at all with public data?** Please assess the two questions separately: whether the arrest comparators and phenotyping support Test A, and whether any available survival endpoint and experimental matching support Test B. A recommendation not to proceed with either test would be useful and will be reflected in the write-up.

Everything is reproducible and the failed validations are public, including the withdrawn circular result: <https://github.com/Jhodytropical/senescence-connectivity-pilot> Run 1 and the corrected report are frozen with checksums. No candidate list is being presented as validated, and no compound here is a recommendation for any use.

Addendum, added 2026-09-29 after the feedback requests were sent

The text above is unchanged, and the exact state linked in those messages is tagged [sent-2026-09-29](#). This note is appended rather than folded in, so nothing shifts under anyone who already read it.

A sharper version of the Test A caveat, which I should have stated outright. Above I wrote that a rapamycin-only comparator risks isolating "rapamycin's specific effects". That undersells the problem:

rapamycin is itself a senomorphic, so senescent-vs-rapamycin-arrested may in practice be senescent-vs-*partially-suppressed-senescent*. The resulting signature would then be biased toward whatever rapamycin does not suppress. Two arrest mechanisms, where one of them is rapamycin, is thinner cover than Q1 implies, and serum withdrawal (GSE162175) carries more of the weight than I gave it. If Test A fails, I would expect this to be why.

This strengthens rather than replaces Q1: is a non-senomorphic second arrest mechanism — contact inhibition, or serum withdrawal alone if its phenotyping holds up — a precondition for the comparison meaning anything?

On stopping. Restating what the section above already commits to: a recommendation not to proceed is a useful answer and will be recorded as the outcome. The risk I am guarding against is building Test A because the pipeline already exists.